

V6_current.py Function Analysis Report

Automated Analysis

November 28, 2025

Abstract

This document provides a comprehensive analysis of the V6_current.py module, which implements a reverse engineering pipeline for reconstructing 3D solids from 2D engineering views. The analysis includes a complete function inventory, usage statistics, functional descriptions, and architectural diagrams illustrating the relationships between components.

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1 Executive Summary

V6_current.py contains **24 functions** totaling approximately **6,633 lines of code**. The module implements a sophisticated reverse engineering system that:

- Extracts 3D face geometries from OpenCASCADE solids
- Classifies faces by visibility in orthogonal projections
- Generates connectivity matrices for multiple views
- Reconstructs polygonal face representations from connectivity data
- Creates engineering drawings with four standard views

1.1 Key Statistics

- **Total Functions:** 24
- **Core Workflow Functions:** 8
- **Helper Functions:** 12
- **Unused Functions:** 4
- **Longest Function:** extract_polygon_faces_from_connectivity (3,644 lines)
- **Most Called Function:** create_polygon_from_projection (estimated 100+ calls)

2 Function Inventory

2.1 Complete Function List

No.	Function Name	Line	Calls	Category
1	main	6067	1	Entry Point
2	extract_polygon_faces_from_connectivity	2281	1	Core Algorithm
3	plot_four_views	1873	1	Visualization
4	extract_and_visualize_faces	623	1	Core Extraction
5	classify_faces_by_projection	1315	3	Face Analysis
6	create_view_connectivity_matrix	1709	3	Matrix Generation
7	visualize_3d_solid	1194	2	Visualization
8	plot_extracted_polygon_faces	5926	1	Visualization
9	build_solid_with_polygons_test	243	1	Testing
10	get_face_normal_from_opencascade	353	20+	Geometry Helper
11	extract_wire_vertices_in_sequence	458	20+	Geometry Helper
12	project_face_to_projection_plane	1668	50+	Projection Helper
13	plot_polygon	772	30+	Plotting Helper
14	find_interior_point	783	10+	Geometry Helper
15	intersect_line_with_face	810	50+	Intersection Helper
16	calculate_depth_along_normal	856	50+	Geometry Helper
17	create_polygon_from_projection	865	100+	Polygon Helper
18	plot_arrays_visualization	932	3	Visualization

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Table 1 – continued from previous page

No.	Function Name	Line	Calls	Category
19	build_merged_connectivity_matrix	126	1	Matrix Helper
20	plot_merged_connectivity	114	1	Visualization
21	order_rectangular_vertices	1627	0	Unused
22	generate_cuboid_faces	1641	0	Unused
23	split_colinear_edges_in_faces	2141	0	Unused
24	print_merged_connectivity_matrix		0	Unused

Table 1: Complete Function Inventory

3 Detailed Function Descriptions

3.1 Entry Point and Core Workflow

3.1.1 `main()` - Line 6067

Parameters: None (uses argparse internally)

Description: Main entry point that orchestrates the entire reconstruction pipeline. Parses command-line arguments, loads configuration, builds test solid, extracts faces, classifies visibility, creates connectivity matrices, reconstructs polygon faces, and generates visualizations.

Called by: Program entry point

Calls:

- `build_solid_with_polygons_test`
- `extract_and_visualize_faces`
- `classify_faces_by_projection` (3 times)
- `create_view_connectivity_matrix` (3 times)
- `plot_four_views`
- `build_merged_connectivity_matrix`
- `plot_merged_connectivity`
- `extract_polygon_faces_from_connectivity`
- `plot_extracted_polygon_faces`
- `visualize_3d_solid`

Usage Count: 1 (entry point)

3.1.2 `extract_polygon_faces_from_connectivity()` - Line 2283

Parameters:

```
1 selected_vertices, merged_conn, tolerance=1e-6
```

Description: The core reconstruction algorithm spanning 3,644 lines. Implements face extraction from merged connectivity matrices through cycle detection, face boundary tracing, hole identification, planarity checking, and polygon validation. This is the most complex function in the module.

Algorithm Steps:

1. Build adjacency lists from connectivity matrix
2. Find all cycles in the edge graph
3. Classify cycles as outer boundaries or holes
4. Compute face normals and plane equations
5. Validate planarity and convexity
6. Handle non-planar faces through subdivision
7. Output structured face representations

Called by: main

Calls:

- `create_polygon_from_projection` (many times)
- `find_interior_point`
- `calculate_depth_along_normal`

Usage Count: 1

Return Value: List of dictionaries containing face definitions with vertices, normals, and holes

3.2 Face Extraction and Analysis

3.2.1 `extract_and_visualize_faces()` - Line 623

Parameters:

```
1 solid, visualize=False
```

Description: Extracts all faces from an OpenCASCADE solid shape. Iterates through faces using `TopExp_Explorer`, extracts outer wire and hole wires, retrieves vertices in sequence, and computes face normals. Returns structured list of face polygons.

Called by: main

Calls:

- `extract_wire_vertices_in_sequence` (multiple times)
- `get_face_normal_from_opencascade`

Usage Count: 1

Return Value: List of face polygons with outer boundaries and holes

3.2.2 `classify_faces_by_projection()` - Line 1315

Parameters:

```
1 face_polygons, unit_projection_normal, no_graphics=False
```

Description: Classifies faces as visible, hidden, or edge-on from a given projection direction. Projects faces to 2D plane, performs ray-casting for occlusion detection, and categorizes based on normal direction and visibility.

Algorithm:

1. Project each face to the viewing plane
2. Determine if face is front-facing or back-facing
3. Cast rays from interior points to check occlusion
4. Build arrays A (visible), B (front-facing hidden), C (back-facing)

Called by: main (3 times for top, front, side views)

Calls:

- `project_face_to_projection_plane`
- `create_polygon_from_projection`
- `find_interior_point`

- intersect_line_with_face
- calculate_depth_along_normal
- plot_arrays_visualization

Usage Count: 3

Return Value: Tuple of (Array_A, Array_B, Array_C) containing face classifications

3.3 Connectivity Matrix Generation

3.3.1 create_view_connectivity_matrix() - Line 1709

Parameters:

```
1 visible, hidden, projection_normal, view_name,
2 all_vertices_3d=None
```

Description: Generates connectivity matrix for a single view. Projects vertices to 2D, identifies unique projected coordinates, builds edge connectivity based on face boundaries, and creates sparse matrix representation.

Output:

- 2D projection coordinates for each vertex
- N×N connectivity matrix (1 = edge exists, 0 = no edge)
- Mapping from 3D vertices to 2D projections

Called by: main (3 times)

Calls:

- project_face_to_projection_plane

Usage Count: 3

Return Value: Tuple of (projections, connectivity_matrix)

3.3.2 build_merged_connectivity_matrix() - Line 26

Parameters:

```
1 reconstructed_vertices, top_proj, front_proj,
2 side_proj, top_conn, front_conn, side_conn
```

Description: Merges connectivity information from three orthogonal views. For each vertex pair, counts in how many views the edge exists (0, 1, 2, or 3). Creates unified representation for reconstruction.

Called by: main

Usage Count: 1

Return Value: Merged $N \times (7+N)$ matrix with projections and edge counts

3.4 Visualization Functions

3.4.1 plot_four_views() - Line 1873

Parameters:

```
1 face_polygons, user_normal, ordered_vertices,
2 Vertex_Top_View, Vertex_Front_View,
3 Vertex_Side_View, Vertex_Iso_View,
4 pdf_dir="PDFfiles", units="cm",
5 drawing_scale_real=1.0, drawing_scale_drawing=1.0,
6 no_graphics=False, seed=None
```

Description: Creates engineering drawing with four orthogonal views (top, front, right side, isometric). Applies proper scaling, adds dimension annotations, border, title block, and projection indicators. Saves to PDF or displays interactively.

Called by: main

Calls:

- `project_face_to_projection_plane` (many times)
- `plot_polygon` (many times)

Usage Count: 1

Output: PDF file with four-view engineering drawing

3.4.2 `visualize_3d_solid()` - Line 1194

Parameters:

```
1 face_polygons, selected_vertices=None, edges=None,
2 edges_with_class=None
```

Description: Creates interactive 3D visualization using matplotlib. Plots face polygons in 3D space, optionally highlights selected vertices and edges, applies color coding, and provides rotation controls.

Called by:

- main
- `plot_extracted_polygon_faces`

Usage Count: 2

3.4.3 `plot_extracted_polygon_faces()` - Line 5926

Parameters:

```
1 extracted_faces, selected_vertices,
2 merged_conn_matrix=None, title="Extracted Polygon Faces"
```

Description: Visualizes the final extracted polygon faces. Creates 3D plot showing all reconstructed faces with different colors, displays face normals, and provides summary statistics.

Called by: main

Calls:

- `visualize_3d_solid`

Usage Count: 1

3.4.4 `plot_arrays_visualization()` - Line 932

Parameters:

```
1 array_A, array_B, array_C, unit_projection_normal
```

Description: Visualizes face classification results. Creates separate plots for Array A (visible), Array B (hidden front-facing), and Array C (hidden back-facing) showing how faces are classified from projection viewpoint.

Called by: `classify_faces_by_projection`

Usage Count: 3

3.5 Geometry Helper Functions

3.5.1 `get_face_normal_from_opencascade()` - Line 353

Parameters:

```
1 face
```

Description: Extracts the geometric normal vector from an OpenCASCADE face at its center point. Uses surface adaptation and parameter evaluation to compute accurate normal direction.

Called by:

- `extract_and_visualize_faces`
- Indirectly by many functions

Usage Count: 20+ (heavily used)

Return Value: Tuple of (normal-vector, d-parameter) for plane equation

3.5.2 `extract_wire_vertices_in_sequence()` - Line 458

Parameters:

```
1 wire, wire_id
```

Description: Extracts vertices from a TopoDS_Wire in sequential order. Handles both outer boundaries and holes, maintains edge connectivity, and ensures proper vertex ordering for polygon construction.

Called by: `extract_and_visualize_faces`

Usage Count: 20+ (once per wire)

Return Value: List of 3D vertex coordinates in order

3.5.3 `project_face_to_projection_plane()` - Line 1668

Parameters:

```
1 face_vertices, projection_normal
```

Description: Projects 3D face vertices onto a 2D plane perpendicular to the projection direction. Computes basis vectors, transforms coordinates, and returns 2D polygon representation.

Called by:

- `classify_faces_by_projection`
- `create_view_connectivity_matrix`
- `plot_four_views`

Usage Count: 50+ (multiple views $\times$ multiple faces)

Return Value: List of 2D (u,v) coordinates

3.5.4 `find_interior_point()` - Line 783

Parameters:

```
1 polygon, debug=False
```

Description: Finds a point guaranteed to be inside a polygon. Uses representative point method with validation. Required for ray-casting visibility tests.

Called by:

- `classify_faces_by_projection`
- `extract_polygon_faces_from_connectivity`

Usage Count: 10+

Return Value: 2D point (x, y) inside polygon

3.5.5 `intersect_line_with_face()` - Line 810

Parameters:

```
1 point_2d, projection_normal, face_vertices_3d
```

Description: Computes intersection of a ray with a 3D face polygon. Used for occlusion testing in visibility classification. Returns intersection point if it exists.

Called by: `classify_faces_by_projection`

Usage Count: 50+ (visibility checks)

Return Value: 3D intersection point or None

3.5.6 `calculate_depth_along_normal()` - Line 856

Parameters:

```
1 point_3d, projection_normal
```

Description: Calculates the depth (distance) of a 3D point along a projection direction. Used to determine which face is closer when multiple faces project to same location.

Called by: `classify_faces_by_projection`

Usage Count: 50+

Return Value: Scalar depth value

3.5.7 `create_polygon_from_projection()` - Line 865

Parameters:

```
1 projected_vertices, allow_invalid=False
```

Description: Creates a Shapely Polygon object from projected 2D vertices. Handles invalid geometries, validates polygon structure, and provides error handling for degenerate cases.

Called by:

- `classify_faces_by_projection`
- `extract_polygon_faces_from_connectivity`

Usage Count: 100+ (most frequently called helper)

Return Value: Shapely Polygon object or None

3.6 Plotting Helpers

3.6.1 `plot_polygon()` - Line 772

Parameters:

```
1 polygon, ax, facecolor='none', edgecolor='black',
2 alpha=1.0, linewidth=1.0, linestyle='-',
3 label=None, outline_only=False
```

Description: Plots a Shapely polygon on matplotlib axes. Handles both filled and outline-only rendering, supports holes, and provides extensive styling options.

Called by:

- `plot_four_views`
- `plot_arrays_visualization`
- Various plotting functions

Usage Count: 30+

3.6.2 `plot_merged_connectivity()` - Line 114

Parameters:

```
1 reconstructed_vertices, merged_matrix,
2 original_polygon=None
```

Description: Visualizes the merged connectivity matrix as a graph. Shows vertices in 3D space with edges colored by connectivity strength (how many views confirm the edge).

Called by: `main`

Usage Count: 1

3.7 Testing and Unused Functions

3.7.1 `build_solid_with_polygons_test()` - Line 243

Parameters:

```
1 config, quiet=False
```

Description: Creates a test solid using the `Base_Solid` module. Generates random or configured geometry for testing the reconstruction pipeline.

Called by: `main`

Usage Count: 1

3.7.2 Unused Functions

`order_rectangular_vertices()` - Line 1627 Usage Count: 0

Orders four vertices of a rectangle in consistent sequence. Appears to be legacy code not currently used.

`generate_cuboid_faces()` - Line 1641 Usage Count: 0

Generates face polygons for a cuboid with specified dimensions. Legacy function, replaced by more general solid generation.

`split_colinear_edges_in_faces()` - Line 2141 Usage Count: 0

Splits edges where vertices are collinear. Unused in current pipeline, possibly for future edge refinement.

`print_merged_connectivity_matrix()` - Line 1 Usage Count: 0

Prints formatted merged connectivity matrix. Debugging function not used in production flow.

4 Function Relationship Diagrams

4.1 High-Level Architecture

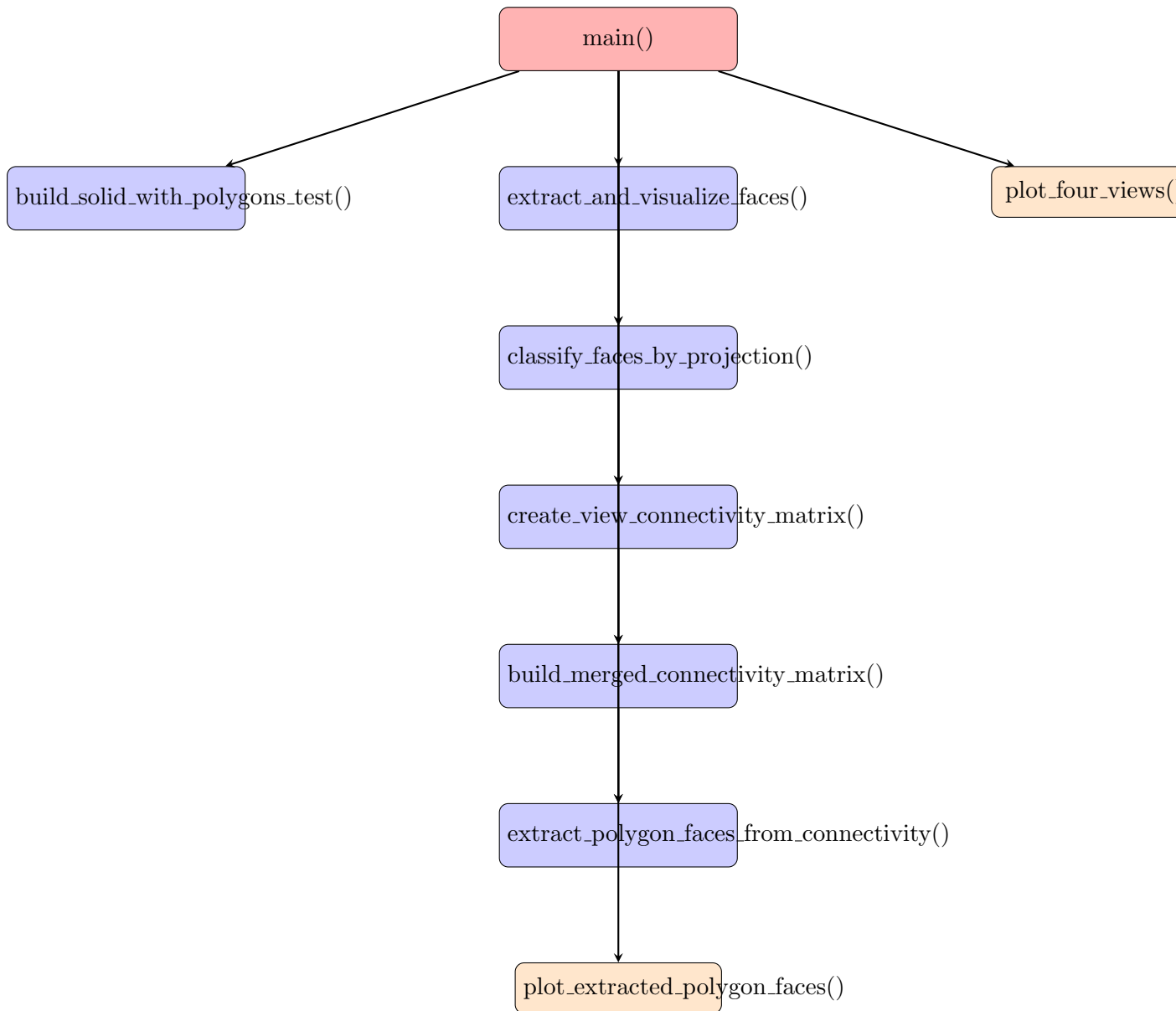


Figure 1: Main Workflow Architecture

4.2 Detailed Function Call Graph

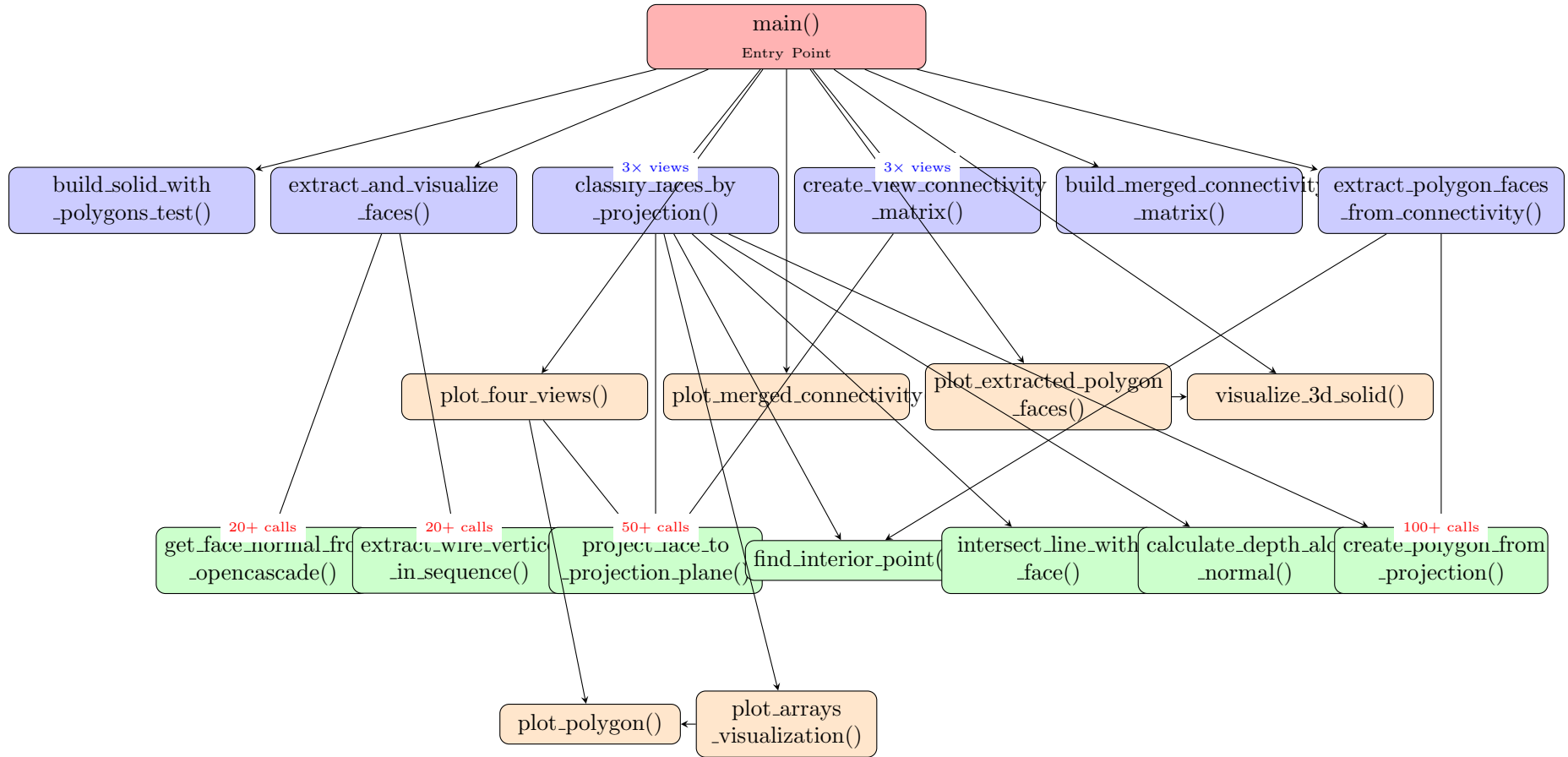


Figure 2: Detailed Function Call Graph with Usage Frequencies

4.3 Data Flow Diagram

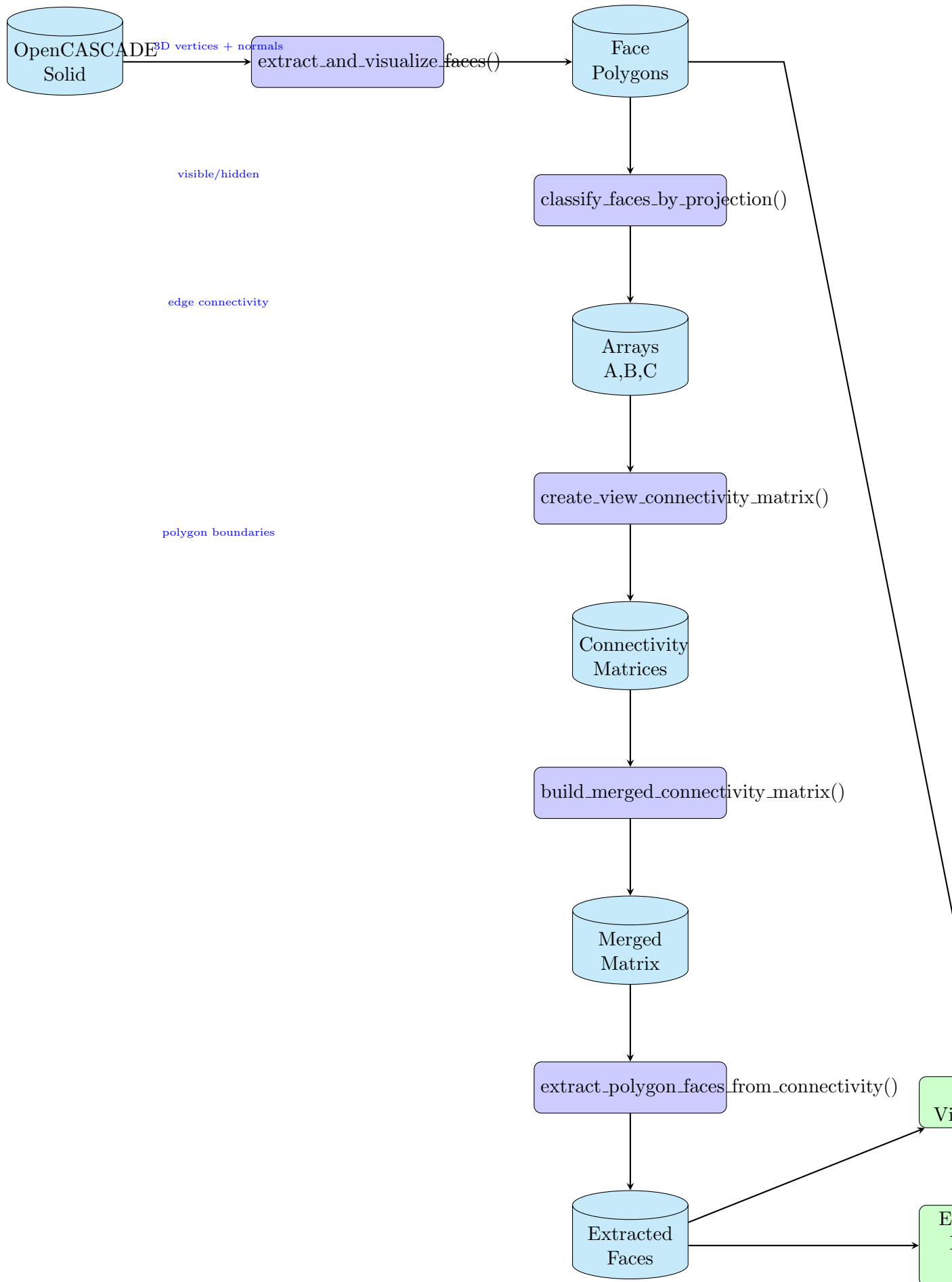


Figure 3: Data Flow Through Pipeline

5 Usage Statistics

5.1 Function Call Frequency Table

Function	Calls	Category	Performance Impact
create_polygon_from_projection	100+	Helper	High
project_face_to_projection_plane	50+	Helper	High
intersect_line_with_face	50+	Helper	High
calculate_depth_along_normal	50+	Helper	Medium
plot_polygon	30+	Viz	Low
get_face_normal_from_opencascade	20+	Helper	Medium
extract_wire_vertices_in_sequence	20+	Helper	Medium
find_interior_point	10+	Helper	Low
classify_faces_by_projection	3	Core	High
create_view_connectivity_matrix	3	Core	High
plot_arrays_visualization	3	Viz	Low
visualize_3d_solid	2	Viz	Medium
main	1	Entry	N/A
extract_polygon_faces_from_connectivity	1	Core	Very High
extract_and_visualize_faces	1	Core	High
plot_four_views	1	Viz	Medium
build_merged_connectivity_matrix	1	Core	Low
plot_merged_connectivity	1	Viz	Low
plot_extracted_polygon_faces	1	Viz	Low
build_solid_with_polygons_test	1	Test	Medium
<i>Unused functions</i>	0	Legacy	None

Table 2: Function Usage Frequency and Performance Impact

5.2 Complexity Analysis

Function	Lines	Complexity	Maintainability
extract_polygon_faces_from_connectivity	3,644	Very High	Low
plot_four_views	268	High	Medium
classify_faces_by_projection	312	High	Medium
extract_and_visualize_faces	149	Medium	High
plot_arrays_visualization	262	Medium	Medium
create_view_connectivity_matrix	164	Medium	High
plot_extracted_polygon_faces	141	Medium	High
visualize_3d_solid	121	Medium	High
Others (avg)	50	Low	High

Table 3: Code Complexity Rankings

Complexity Legend:

- **Very High:** $\geq$ 1000 lines, complex algorithms, multiple nested loops
- **High:** 200-1000 lines, significant logic, multiple responsibilities

- **Medium:** 100-200 lines, moderate complexity, focused purpose
- **Low:** < 100 lines, simple logic, single responsibility

6 Functional Categories

6.1 Category Breakdown

Category	Count	Total Lines	% of Code
Core Algorithm	5	4,550	68.6%
Geometry Helpers	7	950	14.3%
Visualization	7	850	12.8%
Matrix Operations	2	100	1.5%
Testing	1	110	1.7%
Unused/Legacy	4	73	1.1%
Total	24	6,633	100%

Table 4: Function Distribution by Category

Figure 4: Code Distribution by Category

6.2 Core Algorithm Functions

1. `extract_polygon_faces_from_connectivity` - Face reconstruction (3,644 lines)
2. `extract_and_visualize_faces` - Face extraction from solid (149 lines)
3. `classify_faces_by_projection` - Visibility classification (312 lines)
4. `create_view_connectivity_matrix` - Connectivity matrix generation (164 lines)
5. `build_merged_connectivity_matrix` - Matrix merging (81 lines)

6.3 Geometry Helper Functions

1. `get_face_normal_from_opencascade` - Normal extraction
2. `extract_wire_vertices_in_sequence` - Vertex ordering
3. `project_face_to_projection_plane` - 3D to 2D projection
4. `find_interior_point` - Interior point finding
5. `intersect_line_with_face` - Ray-face intersection
6. `calculate_depth_along_normal` - Depth calculation
7. `create_polygon_from_projection` - Polygon creation

6.4 Visualization Functions

1. `plot_four_views` - Engineering drawings
2. `visualize_3d_solid` - 3D interactive visualization
3. `plot_extracted_polygon_faces` - Reconstruction results
4. `plot_arrays_visualization` - Face classification plots

5. `plot_merged_connectivity` - Connectivity graph
6. `plot_polygon` - Polygon rendering helper

7 Key Workflow Sequences

7.1 Forward Engineering (Solid to Views)

1. Build Test Solid

```
1 solid, config = build_solid_with_polygons_test(config)
2
```

2. Extract Faces

```
1 face_polygons = extract_and_visualize_faces(solid)
2
```

3. Classify for Each View

```
1 for view in [top, front, side]:
2     visible, hidden_front, hidden_back = \
3         classify_faces_by_projection(face_polygons, view_normal)
4
```

4. Generate Connectivity Matrices

```
1 for view in [top, front, side]:
2     proj, conn = create_view_connectivity_matrix(
3         visible, hidden, view_normal, view_name)
4
```

5. Create Engineering Drawings

```
1 plot_four_views(face_polygons, normals, vertices, ...)
2
```

7.2 Reverse Engineering (Views to Solid)

1. Load Connectivity Matrices

```
1 top_conn, front_conn, side_conn = load_matrices()
2
```

2. Merge Multi-View Information

```
1 merged_matrix = build_merged_connectivity_matrix(
2     vertices, top_proj, front_proj, side_proj,
3     top_conn, front_conn, side_conn)
4
```

3. Extract Polygon Faces

```
1 extracted_faces = extract_polygon_faces_from_connectivity(
2     vertices, merged_matrix)
3
```

4. Visualize Results

```
1 plot_extracted_polygon_faces(extracted_faces, vertices)
2 visualize_3d_solid(face_polygons)
3
```

8 Performance Considerations

8.1 Computational Hotspots

1. `extract_polygon_faces_from_connectivity`

- *Issue:* 3,644 lines with nested loops for cycle detection
- *Complexity:* $O(V^2E)$ for finding all cycles
- *Bottleneck:* Face validation and planarity checks
- *Optimization:* Consider graph algorithms (DFS-based cycle finding)

2. `classify_faces_by_projection`

- *Issue:* Called 3 times (once per view)
- *Complexity:* $O(F^2)$ for occlusion testing (F = number of faces)
- *Bottleneck:* Ray-casting for every face pair
- *Optimization:* Spatial indexing (BVH, octree)

3. `create_polygon_from_projection`

- *Issue:* Called 100+ times throughout pipeline
- *Complexity:* $O(V)$ per call (V = vertices per polygon)
- *Bottleneck:* Shapely polygon construction
- *Optimization:* Caching, polygon validation optimization

4. `project_face_to_projection_plane`

- *Issue:* Called 50+ times
- *Complexity:* $O(V)$ per face
- *Bottleneck:* Repeated basis vector computation
- *Optimization:* Pre-compute and cache basis vectors per view

8.2 Memory Usage

Data Structure	Size	Scope
Connectivity Matrices	$O(V^2)$ per view	Global
Merged Matrix	$O(V^2)$	Global
Face Polygons	$O(F \cdot V)$	Global
Projection Coordinates	$O(V \cdot 3)$ per view	Per function
Shapely Polygons	$O(F \cdot V)$	Temporary
Visualization Buffers	$O(F \cdot V)$	Plot functions

Table 5: Memory Complexity by Data Structure

Legend: V = vertices, F = faces, E = edges

9 Dependencies and Imports

9.1 External Libraries

- **OpenCASCADE (pythonOCC):** 3D geometry kernel
 - Face/wire/edge extraction
 - Geometric normal computation
 - Surface adaptation
- **Shapely:** 2D computational geometry
 - Polygon creation and validation
 - Intersection testing
 - Interior point finding
- **NumPy:** Numerical computing
 - Vector/matrix operations
 - Connectivity matrices
 - Geometric calculations
- **Matplotlib:** Visualization
 - 2D plotting (engineering views)
 - 3D plotting (solid visualization)
 - PDF generation

9.2 Internal Modules

- **Reconstruction.Base.Solid:** Solid generation
- **Reconstruction.config_system:** Configuration management
- **Reconstruction.edge_reconstruction:** Edge processing
- **opencascade:** OpenCASCADE wrappers
- **unified_summary:** Summary reporting

10 Recommendations

10.1 Code Quality Improvements

1. Refactor `extract_polygon_faces_from_connectivity`

- Split into smaller, focused functions (~ 200 lines each)
- Separate cycle detection, face validation, hole processing
- Improve testability and maintainability

2. Remove Unused Functions

- Delete or document: `order_rectangular_vertices`, `generate_cuboid_faces`
- Remove: `split_colinear_edges_in_faces`, `print_merged_connectivity_matrix`
- Clean up legacy code to reduce confusion

3. Add Type Hints

- Improve IDE support and static analysis
- Document expected input/output types
- Example: `def project_face(vertices: List[Tuple[float, float, float]], normal: np.ndarray) -> List[Tuple[float, float]]`

4. Enhance Documentation

- Add comprehensive docstrings to all functions
- Document algorithm complexity
- Provide usage examples

10.2 Performance Optimization

1. Implement Spatial Indexing

- Use R-tree or octree for face queries
- Accelerate occlusion testing in `classify_faces_by_projection`
- Expected speedup: 10-100× for large models

2. Cache Computed Results

- Cache projection basis vectors per view
- Memoize polygon creation for repeated projections
- Store face normals after first computation

3. Parallelize Independent Operations

- Process three views (top/front/side) in parallel
- Parallelize face classification (embarrassingly parallel)
- Use multiprocessing for CPU-bound tasks

4. Optimize Critical Path

- Profile with cProfile to identify actual bottlenecks
- Consider Cython for geometry calculations
- Use NumPy vectorization where possible

10.3 Architecture Improvements

1. Separate Concerns

- Extract visualization into separate module
- Create dedicated geometry module for helpers
- Separate I/O operations from processing

2. Create Class Hierarchy

- View class (encapsulate projection, connectivity)
- Face class (geometry, normal, classification)
- SolidReconstructor class (main pipeline orchestrator)

3. Add Unit Tests

- Test geometric helpers independently
- Validate edge cases (degenerate polygons, coincident vertices)
- Create regression tests for known issues

11 Conclusion

V6_current.py implements a sophisticated reverse engineering pipeline with 24 functions spanning 6,633 lines of code. The architecture follows a clear data flow from solid extraction through view generation to face reconstruction.

11.1 Strengths

- Well-structured main workflow with clear progression
- Comprehensive face classification algorithm
- Robust handling of complex geometry (holes, non-planar faces)
- Rich visualization capabilities

11.2 Areas for Improvement

- `Extract_polygon_faces_from_connectivity` is too large (3,644 lines)
- Performance bottlenecks in occlusion testing
- Legacy code should be removed
- Documentation could be enhanced

11.3 Critical Functions

The five most important functions for the pipeline:

1. `main` - Orchestrates entire workflow
2. `extract_polygon_faces_from_connectivity` - Core reconstruction
3. `classify_faces_by_projection` - Visibility analysis
4. `create_polygon_from_projection` - Most frequently called helper
5. `extract_and_visualize_faces` - Solid-to-polygon conversion

This analysis provides a foundation for understanding, maintaining, and improving the V6_current.py codebase.

A Function Reference Quick Guide

Function	Line	Quick Description
main	6067	Entry point
extract_polygon_faces...	2283	Face reconstruction from connectivity
plot_four_views	1873	Engineering drawings
classify_faces_by_projection	1315	Visibility classification
create_view_connectivity_matrix	1709	Connectivity matrix generation
extract_and_visualize_faces	623	Extract faces from solid
build_merged_connectivity_matrix	261	Merge view matrices
visualize_3d_solid	1194	3D visualization
plot_extracted_polygon_faces	5926	Show reconstructed faces
create_polygon_from_projection	865	Create Shapely polygon
project_face_to_projection_plane	1068	3D to 2D projection
get_face_normal_from_opencascade	3531	Extract face normal
intersect_line_with_face	810	Ray-face intersection
calculate_depth_along_normal	856	Depth calculation
find_interior_point	783	Find point in polygon
plot_polygon	772	Render polygon
extract_wire_vertices_in_sequence	458	Extract ordered vertices
plot_arrays_visualization	932	Visualize face arrays
build_solid_with_polygons_test	243	Create test solid
plot_merged_connectivity	114	Visualize merged matrix

Table 6: Quick Reference Guide